

README

OpenDesign — Supervised Bring-Up

Constitution: §11.4.162 (OpenDesign is a mandatory core dependency for every user-facing UI surface) · §11.4.161 (rootless — no root, no system service) · §11.4.133 (host safety) · §11.4.77 (gitignored-artifact re-obtain mechanism) · §11.4.174 (process-ownership verification before inspecting/acting on any process).

What this replaces

An earlier stream (scratchpad/phase1_opendesign_install.md) provisioned the OpenDesign daemon (nexu-io/open-design) as a bare **detached process**: clone → pnpm install → nohup ... & with no health supervision, no restart-on-failure, and its pidfile/log living *outside* the tracked tree (in the sibling .opendesign-src / clone). That was sufficient to prove the integration works end-to-end but is not a managed service — a crashed or hung daemon just stays down/hung until a human notices.

bringup.sh is the supervised replacement: **idempotent, health- checked, restarts only a positively-verified-ours process**, and its own pidfile/log live inside this repo (gitignored) so they are discoverable without knowing the external clone path.

Usage

```
scripts/opendesign/bringup.sh           # same as `up`
scripts/opendesign/bringup.sh up         # ensure a healthy
    daemon is running
scripts/opendesign/bringup.sh status     # report state only, no
    side effects
scripts/opendesign/bringup.sh stop       # stop it, only if
    verified ours
scripts/opendesign/bringup.sh restart    # stop (if ours) + up
```

up (the default) is safe to run any number of times:

- **Already healthy** → prints the health probe result and exits 0 (no-op — proven by re-running it twice in a row, see evidence below).
- **Port occupied by our own daemon but failing health checks** → restarts it (SIGTERM, wait, SIGKILL fallback, then a fresh start).

- **Port occupied by something that does NOT match our daemon's cmdline signature** (apps/daemon/dist/cli.js) → refuses to touch it and exits non-zero (§11.4.174 — never kill an unverified process; §11.4.122 — never silently remove something we didn't launch).
- **Nothing running** → provisions the source if missing, installs deps if missing, starts the daemon, and polls its health endpoint for up to OD_HEALTH_TIMEOUT_S (default 180s).

Health check

GET http://<OD_HOST>:<OD_PORT>/api/health → {"ok":true,"version":"X.Y.Z"}. The script parses this JSON with jq — a bare HTTP 200 is not trusted on its own (§11.4.6).

Re-obtain mechanism (§11.4.77)

The daemon's source is **not vendored into this repo** — it is a ~1.9 GB checkout of nexu-io/open-design, excluded per §11.4.30. This script is the documented, automated regeneration mechanism: on any fresh host with no prior checkout, bringup.sh up will:

1. `git clone git@github.com:nexu-io/open-design.git` (SSH, Rule 3) into OD_SRC_DIR (default: a sibling directory of the repo root, <parent-of-repo-root>/opendesign-src/open-design — kept OUTSIDE the tracked tree, matching the prior stream's layout).
2. `corepack enable && pnpm install` (bounded by OD_INSTALL_TIMEOUT_S, default 600s) if node_modules/ or the built daemon (apps/daemon/dist/) is missing.
3. Start the daemon directly via its stable entry point, `node apps/daemon/dist/cli.js --port <OD_PORT> --host <OD_HOST> --no-open` — **not** the upstream dev launcher `pnpm tools-dev run web`, which deliberately probes for a free port and ignores --port/OD_PORT, making it unsuitable for a stable, committed :7456 integration (the .mcp.json open-design MCP entry is pinned to OD_DAEMON_URL=http://localhost:7456).

Toolchain requirement: Node.js ~24, pnpm 10.33.x (matches upstream's pinned quickstart). `corepack enable` activates the pinned pnpm version from the checkout's package.json.

Config (env-overridable, project-agnostic per §11.4.28)

Var	Default
OD_PORT	7456
OD_HOST	127.0.0.1
OD_SRC_DIR	<parent-of-repo-root>/opendesign-src/ open-design
OD_REPO_SSH	git@github.com:nexu-io/open-design.git
OD_HEALTH_TIMEOUT_S	180
OD_INSTALL_TIMEOUT_S	600

No operator home-directory path is hardcoded — the source directory default is derived from the repo root (`$(dirname "$REPO_ROOT")`), and every value above may be overridden.

Runtime artifacts (gitignored)

- `scripts/opendesign/run/daemon.pid` — the PID this script launched.
- `scripts/opendesign/run/daemon.log` — the daemon's stdout/stderr.
- `scripts/opendesign/run/bringup.log` — this script's own supervisor log (every action + health-probe result, timestamped).

Already covered by the repo's global `*.pid / *.log .gitignore` patterns; `scripts/opendesign/run/` is also listed explicitly for defense-in-depth (§11.4.53).

Honest limitations (§11.4.6)

- This is a **user-level supervised script**, not an OS-managed service (no systemd unit, no launchd agent) — per the task's explicit constraint (§11.4.133 host safety / §11.4.161 rootless: no root, no system-wide install). It does **not** survive a host reboot on its own; re-run `bringup.sh` up after a reboot (or wire it into whatever user-level session-start mechanism the operator already uses — out of scope here since that would require assumptions about the host's init system).
- Restart only ever targets a process whose `/proc/<pid>/cmdline` matches the daemon's own entry point (`apps/daemon/dist/cli.js`). A foreign process squatting on the configured port is left alone and the script exits non-zero with a clear message (§11.4.174).

- `od_generate_design` (BYOK-gated) is unaffected by this script — it still needs operator secrets in `.env` per `docs/OPENDESIGN.md`; `bringup.sh` only supervises the daemon process, not BYOK configuration.

Evidence

See `scratchpad/opendesign_evidence/bringup_supervised_*.log` for a captured run proving: stop (ownership-verified) → confirmed port-free → status reporting UNHEALTHY/DOWN → up bringing the daemon back healthy in ~3s (source already provisioned) → a second up call no-op'ing (idempotency) → the daemon's SQLite-persisted project data (`helixcode-brand`) surviving the restart.